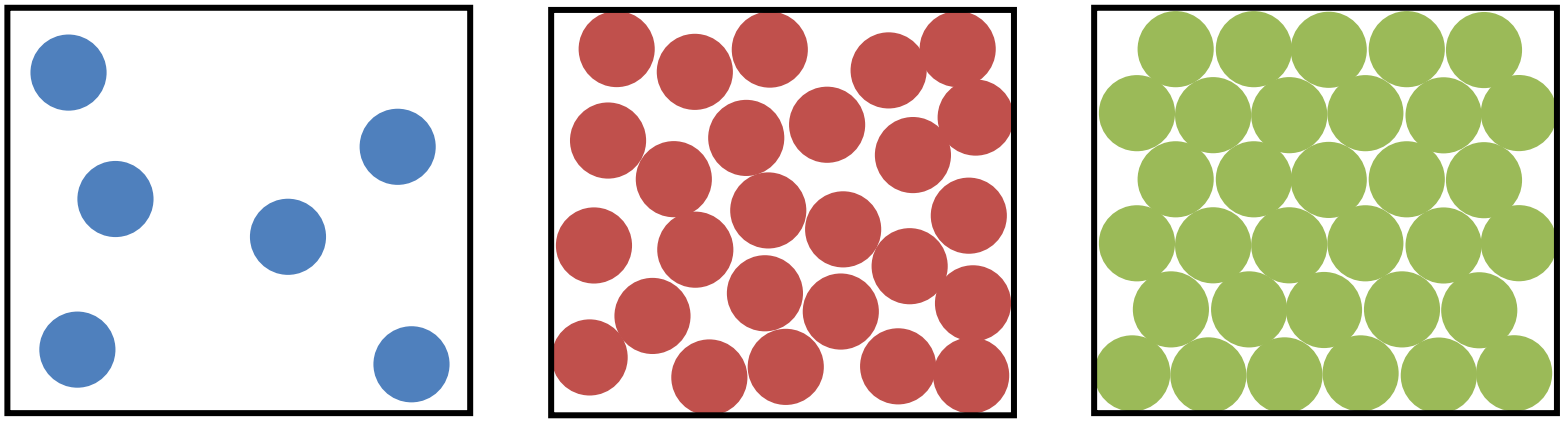
Physical Chemistry of Materials

Lecture 10

Solids

Gas, Liquid, and Solid

condensed phase



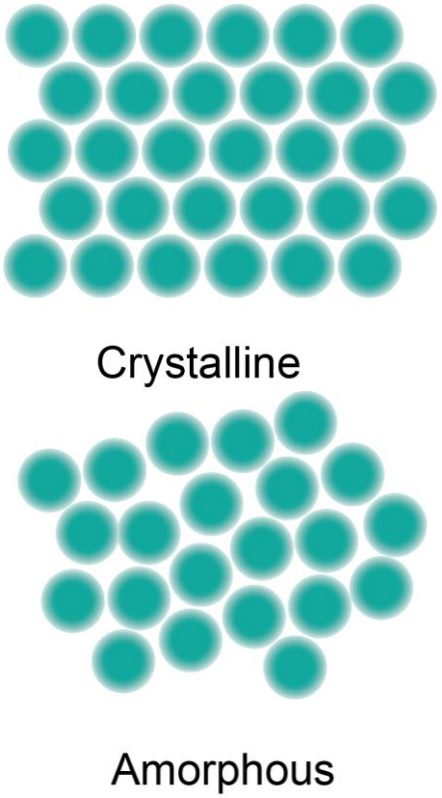
Gas phase: kinetic energy $\gg$ potential energy

Condensed phase: kinetic energy $\lessapprox$ potential energy

$\frac{1}{2}mv^2$ various forms

Condensed Phases

- Crystalline solids
- Amorphous solids
- Liquids (cf. glasses)
- “Soft matters”
  - Polymers
  - Colloids
  - Liquid crystals …



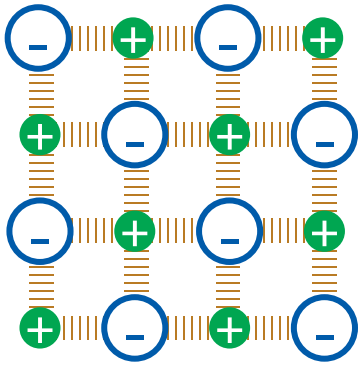
(Image: https://commons.wikimedia.org/wiki/File:Crystalline_vs._Amorphous_solid.png)

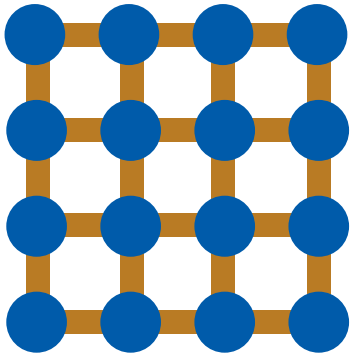
소재물리화학의 구성

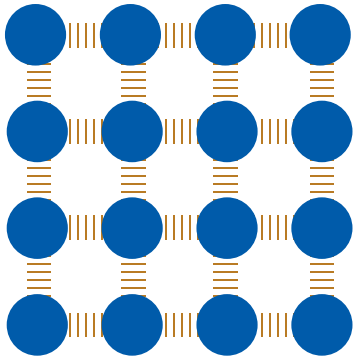
고전 열역학	고체	연성물질
열역학 복습	고체의 구조	액체
상 전이	고체의 진동	유리
통계역학	고체의 전기적 성질	고분자의 구조
기초 이론	고체의 자기적 성질	고분자 용액
이상 기체	표면화학	고분자의 점탄성
상호작용 계	반도체, 나노화학, …	생체 고분자

Bonding in Solids

- Ionic solids: ionic bonds
- Covalent solids: covalent bonds
- Molecular solids: intermolecular interactions

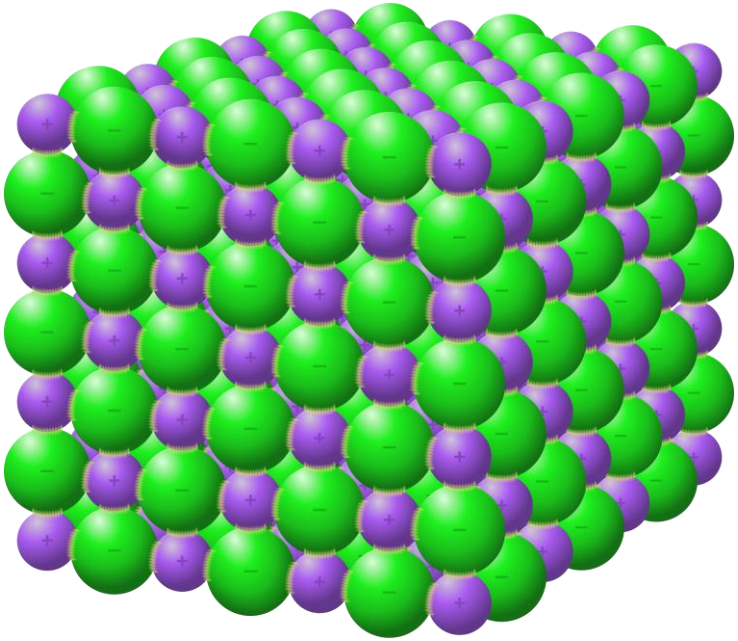






Metals need more discussion

Ionic Solids



Sodium chloride (NaCl)

(Image: https://commons.wikimedia.org/wiki/File:NaCl_bonds.svg)

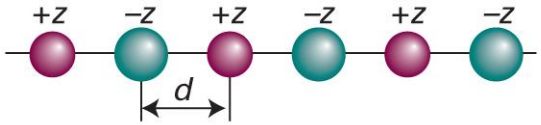
Lattice Energy of Ionic Solids

- Lattice energy:

$$\Delta U \text{ for } MX(s) \longrightarrow M^{+}(g) + X^{-}(g)$$

- Two main contributions
  - Electrostatic interactions (attraction + repulsion)
  - Pauli repulsion

1-Dimensional Model



$$E_{\mathrm{p}} = \frac{1}{4\pi\varepsilon_0}\left(-\frac{z^2e^2}{d} + \frac{z^2e^2}{2d} - \frac{z^2e^2}{3d} + \frac{z^2e^2}{4d} + \cdots\right)$$

$$= \frac{z^2e^2}{4\pi\varepsilon_0 d}\left(-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \cdots\right) = -\frac{z^2e^2}{4\pi\varepsilon_0 d} \times \ln 2$$

Consider two sides for each ion:

$$E_{\mathrm{p}} = -2\ln 2 \times \frac{z^2e^2}{4\pi\varepsilon_0 d}$$

Generalization

$$E_{\mathrm{p}} = -2 \ln 2 \times \frac{z^2 e^2}{4\pi\varepsilon_0 d}$$

For ions with arbitrary charges z_{A} and z_{B} in 3 dimensions,

$$E_{\mathrm{p}} = -A \times \frac{|z_{\mathrm{A}} z_{\mathrm{B}}| e^2}{4\pi\varepsilon_0 d}$$

where A is the Madelung constant (structure-dependent).

Pauli Repulsion

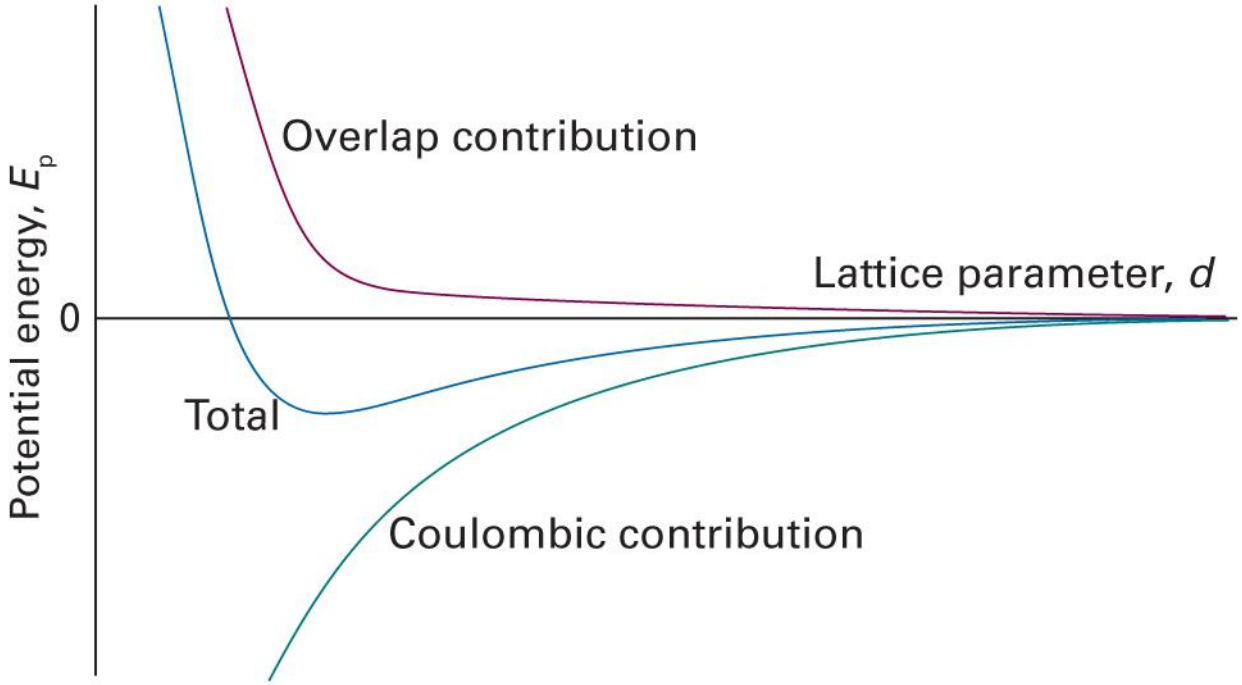
modeled as an exponential potential (lecture 07)

$$E_{\mathrm{p}}^{*} = C e^{-d/d^{*}}$$

- Total potential energy $E_{\mathrm{p,tot}} = E_{\mathrm{p}} + E_{\mathrm{p}}^{*}$

$$E_{\mathrm{p,tot}} = -A \frac{|z_{\mathrm{A}} z_{\mathrm{B}}| e^{2}}{4\pi\varepsilon_{0} d} + C e^{-d/d^{*}}$$

Total Potential Energy



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 15C.13)

Determination of d_{eq}

$$E_{\mathrm{p,tot}}(d) = -A\frac{|z_{\mathrm{A}}z_{\mathrm{B}}|e^2}{4\pi\varepsilon_0 d} + Ce^{-d/d^*}$$

The structure is stabilized when $E_{\mathrm{p,tot}}$ is minimal:

$$\frac{\mathrm{d}E_{\mathrm{p,tot}}}{\mathrm{d}d} = A\frac{|z_{\mathrm{A}}z_{\mathrm{B}}|e^2}{4\pi\varepsilon_0 d_{\mathrm{eq}}^2} - \frac{C}{d^*}e^{-d_{\mathrm{eq}}/d^*} = 0$$

$$Ad^*\frac{|z_{\mathrm{A}}z_{\mathrm{B}}|e^2}{4\pi\varepsilon_0 d_{\mathrm{eq}}^2} = Ce^{-d_{\mathrm{eq}}/d^*}$$

Born-Mayer Equation

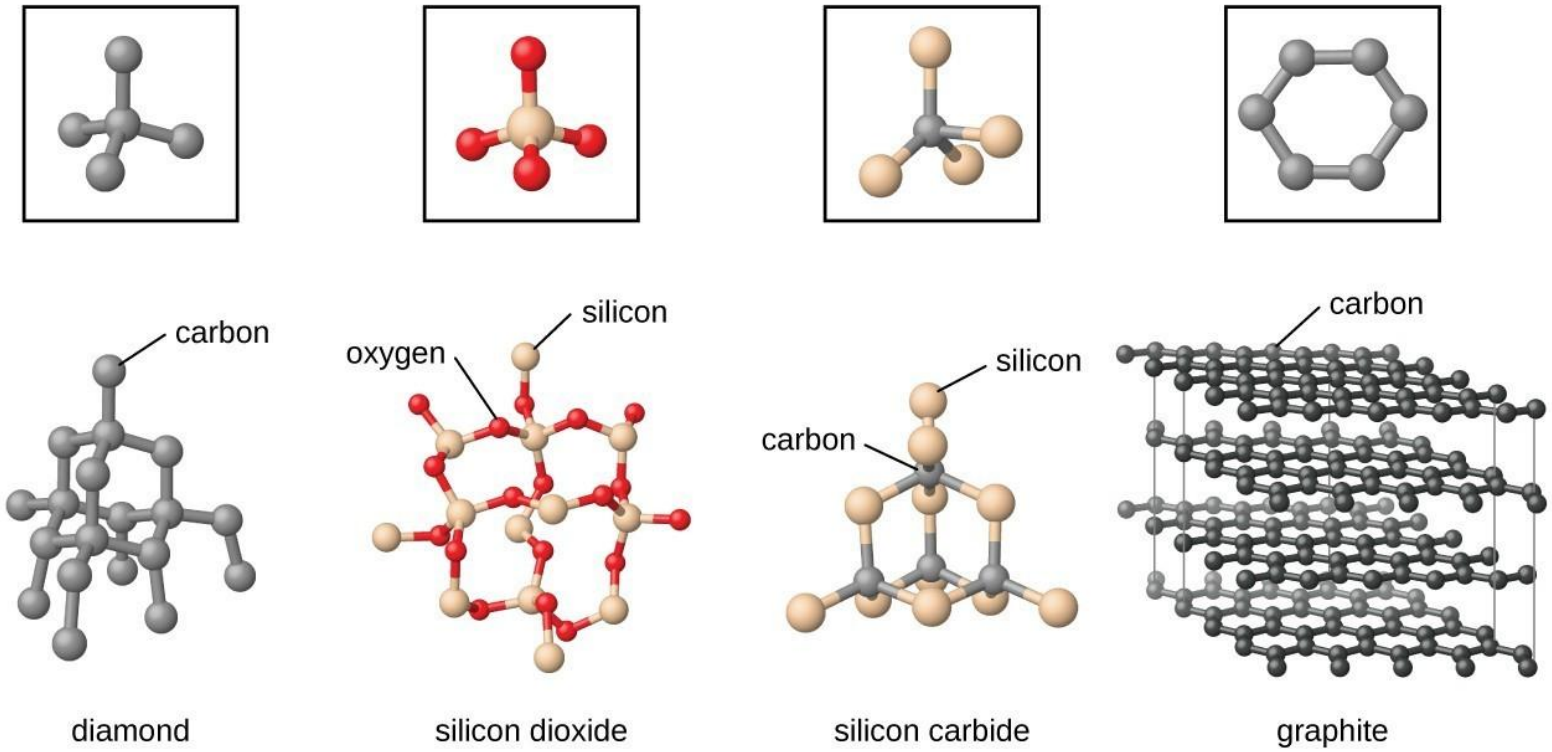
$$Ad^* \frac{|z_A z_B| e^2}{4\pi\varepsilon_0 d_{eq}^2} = Ce^{-d_{eq}/d^*}$$

$$E_{p,tot}(d_{eq}) = -A \frac{|z_A z_B| e^2}{4\pi\varepsilon_0 d_{eq}} + Ce^{-d_{eq}/d^*}$$

$$= -A \frac{|z_A z_B| e^2}{4\pi\varepsilon_0 d_{eq}} + Ad^* \frac{|z_A z_B| e^2}{4\pi\varepsilon_0 d_{eq}^2}$$

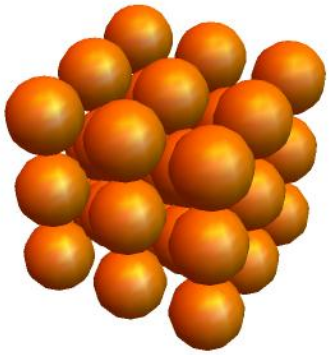
$$E_{p,tot}(d_{eq}) = -A \frac{|z_A z_B| e^2}{4\pi\varepsilon_0 d_{eq}} \left(1 - \frac{d^*}{d_{eq}}\right)$$

Covalent Solids

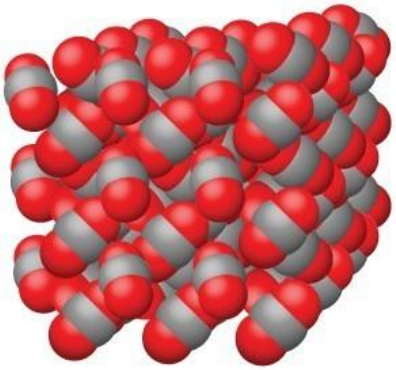


(Image: https://courses.lumenlearning.com/chemistryformajors/chapter/the-solid-state-of-matter-2/)

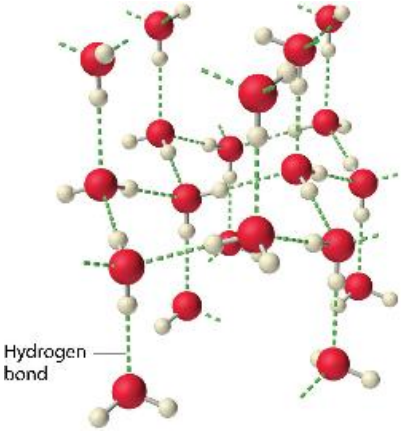
Molecular Solids



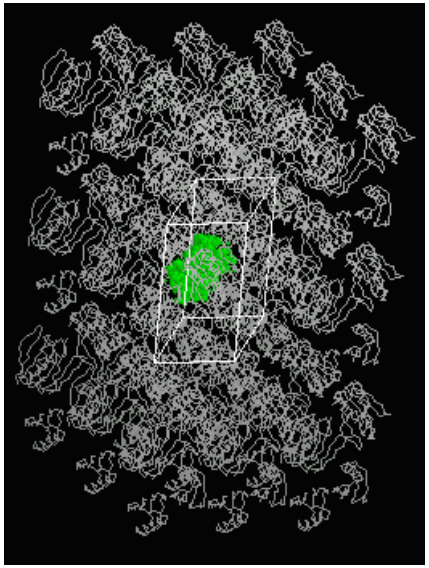
argon



carbon dioxide



water

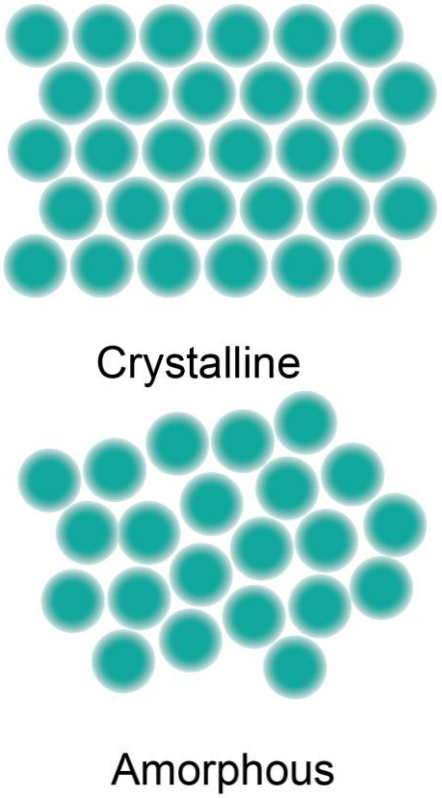


protein

(Images: https://demonstrations.wolfram.com/BasicStructurePlaneOrientationAndFacettingInTheFaceCenteredA/
https://courses.lumenlearning.com/chemistryformajors/chapter/the-solid-state-of-matter-2/
https://thefactfactor.com/facts/pure_science/chemistry/physical-chemistry/classification-of-crystalline-solids/6939/
https://www.umass.edu/molvis/decatur/pe2.727/protexpl/xtlcon.htm)

Condensed Phases

- Crystalline solids
- Amorphous solids
- Liquids (cf. glasses)
- “Soft matters”
  - Polymers
  - Colloids
  - Liquid crystals …



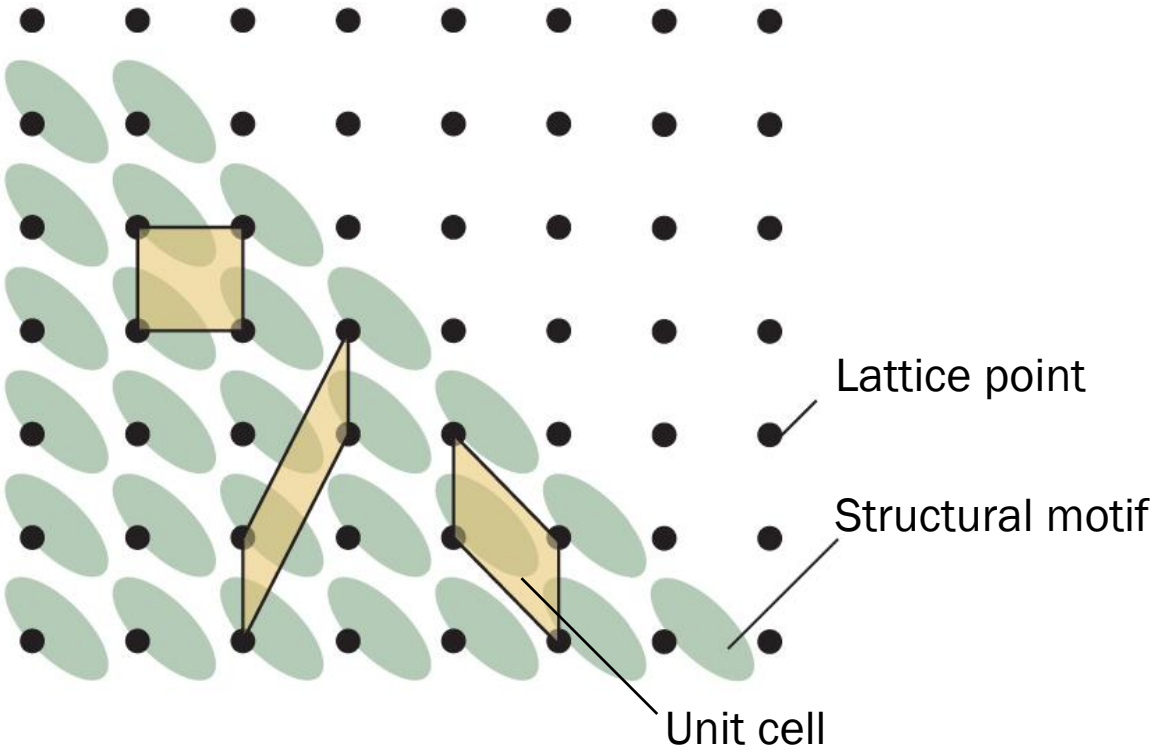
(Image: https://commons.wikimedia.org/wiki/File:Crystalline_vs._Amorphous_solid.png)

Crystalline Solids

- Perfect crystals: built up from regularly repeating "structural motifs"
- Real crystals can contain defects

Unit Cells

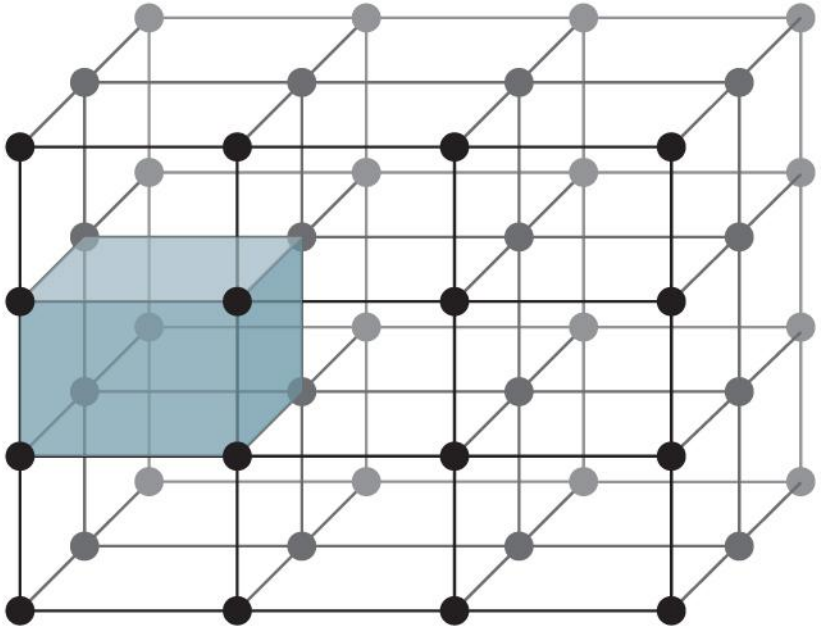
whose purely translational displacements can span the entire space



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figures 15A.1 and 15A.2)

Primitive Unit Cell

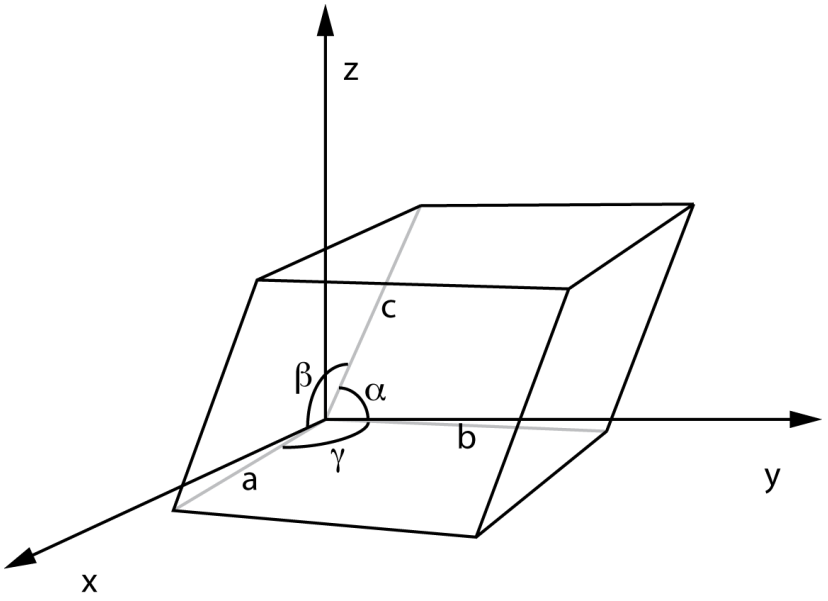
contains only one lattice point



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 15A.3)

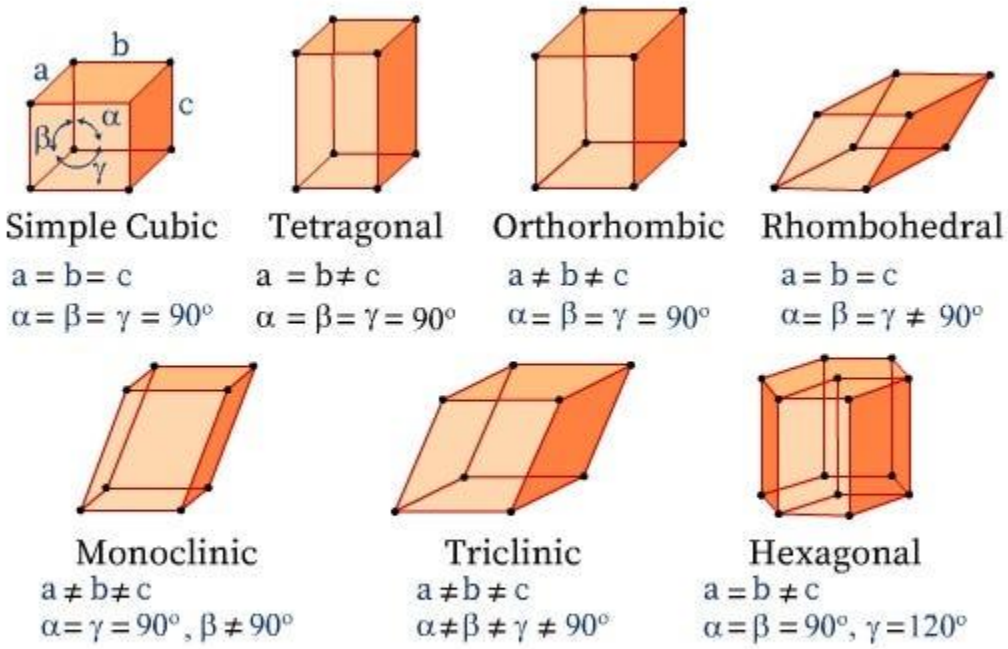
Lattice Constants

- Lengths of sides: a, b, c
- Angles between sides: α, β, γ



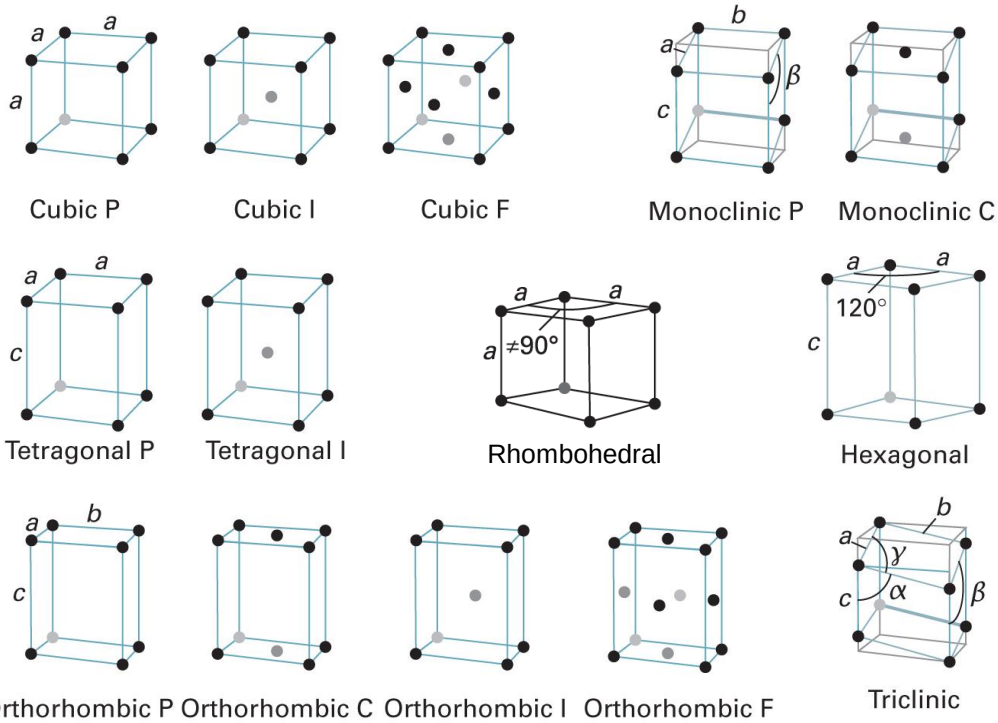
(Image: https://commons.wikimedia.org/wiki/File:UnitCell.png)

Lattice Systems



(Image: https://medium.com/@melyne.zhou/visualizing-crystals-with-vesta-unraveling-crystallography-fe3915a2d6eb)

14 Bravais Lattices

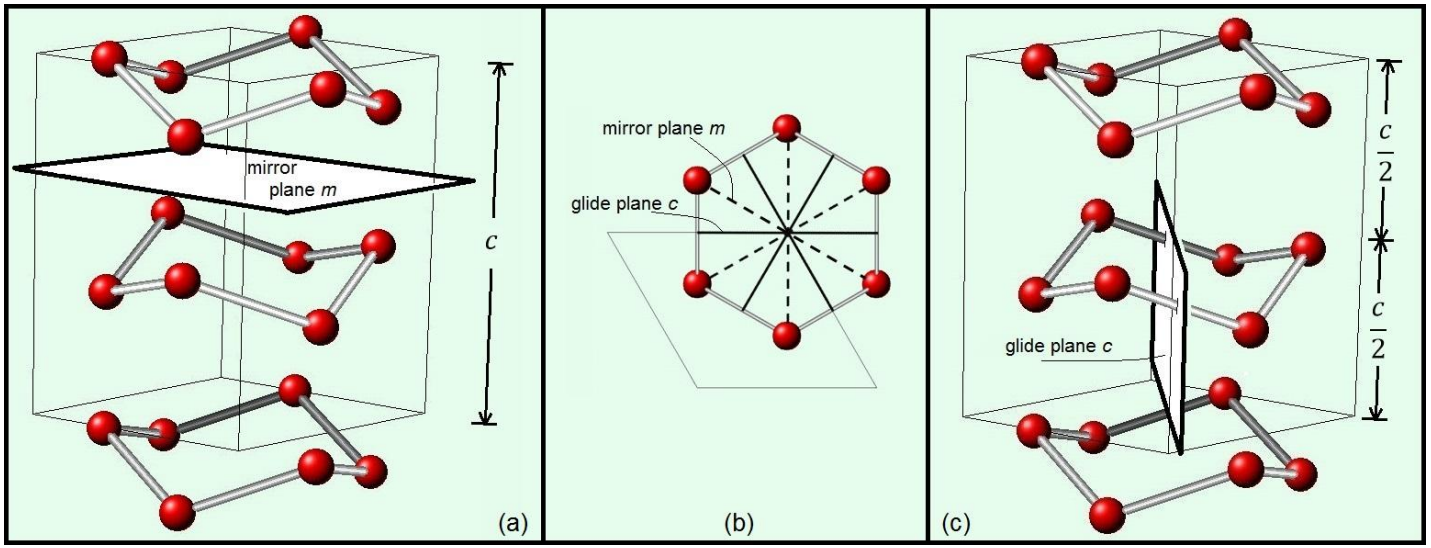


(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 15A.8)

Crystal Symmetry

lattice: hexagonal

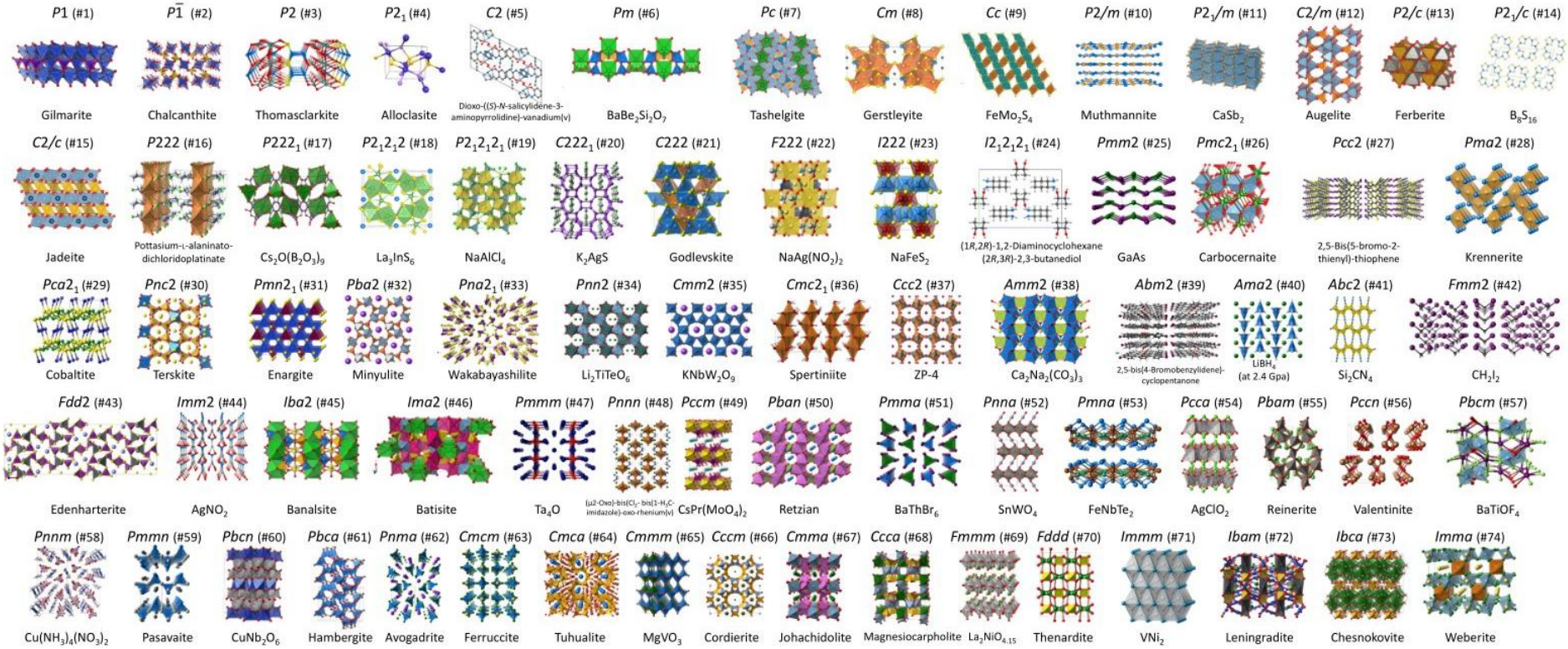
Ice I_h system



lattice symmetry ≠ crystal symmetry

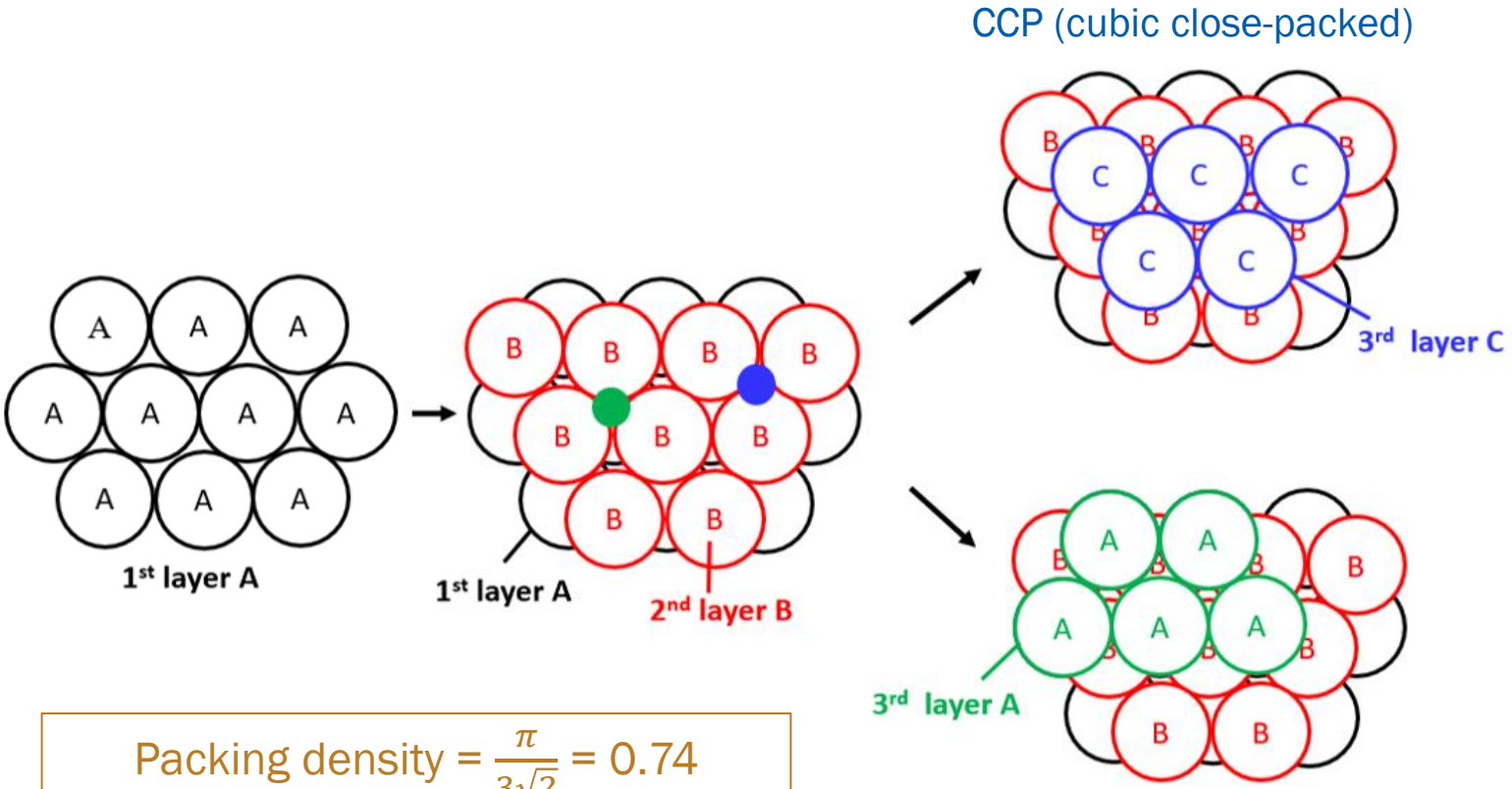
(Image: https://commons.wikimedia.org/wiki/File:Ice_Ih_Space_Group.jpg)

230 Space Groups



(Image: https://crystalsymmetry.wordpress.com/)

Close Packing

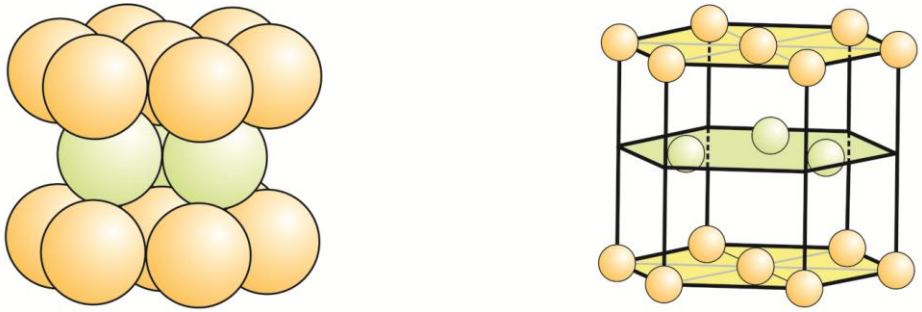


Packing density = $\frac{\pi}{3\sqrt{2}}$ = 0.74

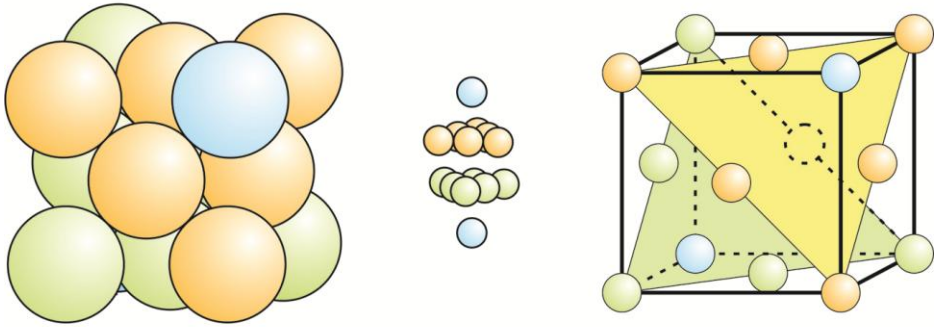
(Kepler conjecture: proven in 1998)

(Image: https://commons.wikimedia.org/wiki/File:Cubic_Closest_Packing_(CCP)_and_Hexagonal_Closet_Packing_(HCP).png)

Unit Cells of CCP and HCP



a. hexagonal closest packing



b. cubic closest packing

(Image: https://opengeology.org/Mineralogy/13-crystal-structures/)

Crystal Structures of Elements at STP

STP - Standard Temperature and Pressure

BCC - Body-centered Cubic
FCC - Face-centered Cubic
HEX - Simple Hexagonal
HCP - Close-packed Hexagonal
DHCP - Double Close-packed Hexagonal
RHO - Rhombohedral

BCT - Body-centered Tetragonal
ORTH - Orthorhombic
DC - Diamond Cubic
DT - Diamond Tetragonal
SC - Simple Cubic
* predicted crystal structure

H HEX																He HCP
Li BCC	Be HCP											B RHO	C HEX	N complex HCP	O P-cubic	F P-cubic	Ne FCC
Na BCC	Mg HCP											Al FCC	Si DC	P ORTH	S ORTH	Cl complex C-ORTH	Ar FCC
K BCC	Ca FCC	Sc HCP	Ti HCP	V BCC	Cr BCC	Mn α-Mn	Fe BCC	Co HCP	Ni FCC	Cu FCC	Zn HCP	Ga complex F-ORTH	Ge DC	As P-RHO	Se complex HEX	Br complex C-ORTH	Kr FCC
Rb BCC	Sr FCC	Y HCP	Zr HCP	Nb BCC	Mo BCC	Tc HCP	Ru HCP	Rh FCC	Pd FCC	Ag FCC	Cd HCP	In BCT	Sn DT	Sb P-RHO	Te complex HEX	I complex C-ORTH	Xe FCC
Cs BCC	Ba BCC	57-71	Hf HCP	Ta BCC	W BCC	Re HCP	Os HCP	Ir FCC	Pt FCC	Au FCC	Hg RHO	Tl HCP	Pb FCC	Bi RHO	Po SC	At FCC*	Rn FCC*
Fr BCC*	Ra BCC	89-103	Rf HCP*	Db BCC*	Sg BCC*	Bh HCP*	Hs HCP*	Mt FCC*	Ds BCC*	Rg BCC*	Cn HCP*	Nh HCP*	Fl FCC*	Mc UNKNOWN	Lv UNKNOWN	Ts UNKNOWN	Og FCC*

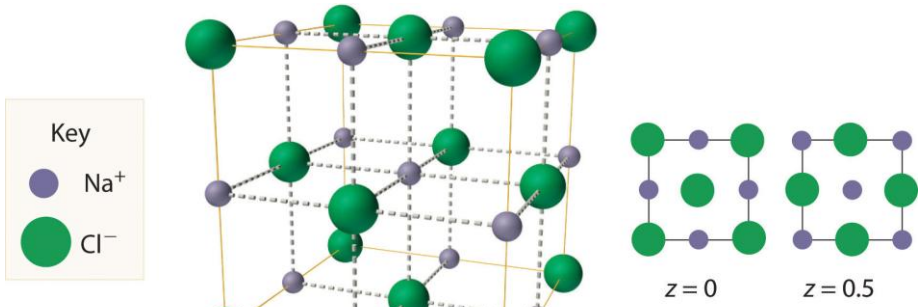
Solid state at STP
Liquid state at STP
Gaseous state at STP

La DHCP	Ce DHCP	Pr DHCP	Nd DHCP	Pm DHCP	Sm complex RHO	Eu BCC	Gd HCP	Tb HCP	Dy HCP	Ho HCP	Er HCP	Tm HCP	Yb FCC	Lu HCP
Ac FCC	Th FCC	Pa BCT	U ORTH	Np ORTH	Pu MONO	Am DHCP	Cm DHCP	Bk DHCP	Cf DHCP	Es FCC	Fm FCC*	Md FCC*	No FCC*	Lr HCP*

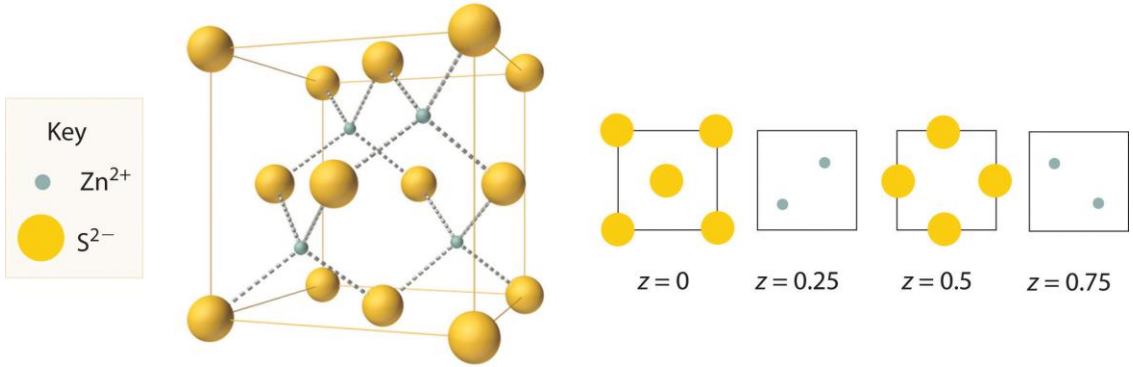
(Image: https://msestudent.com/what-are-crystals-and-grains/)

Binary Compounds

Rock-salt structure

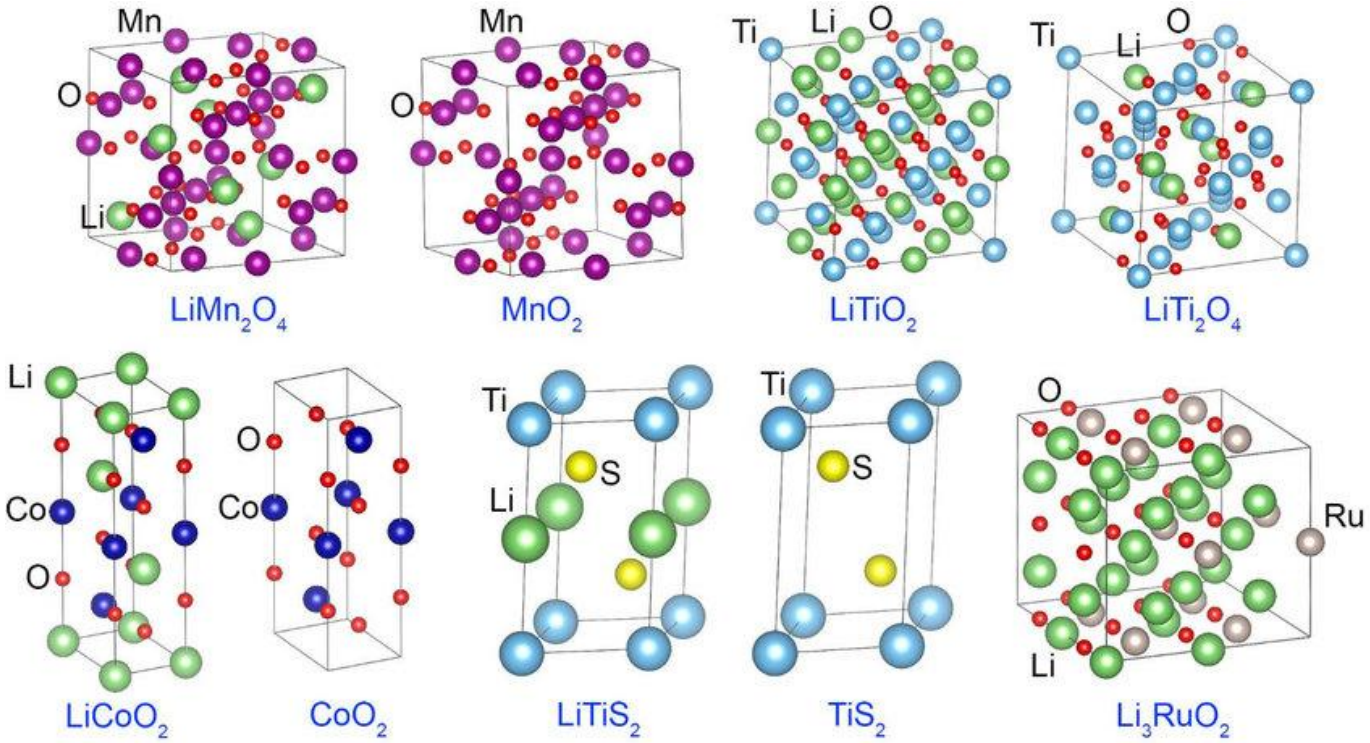


zinc-blend structure



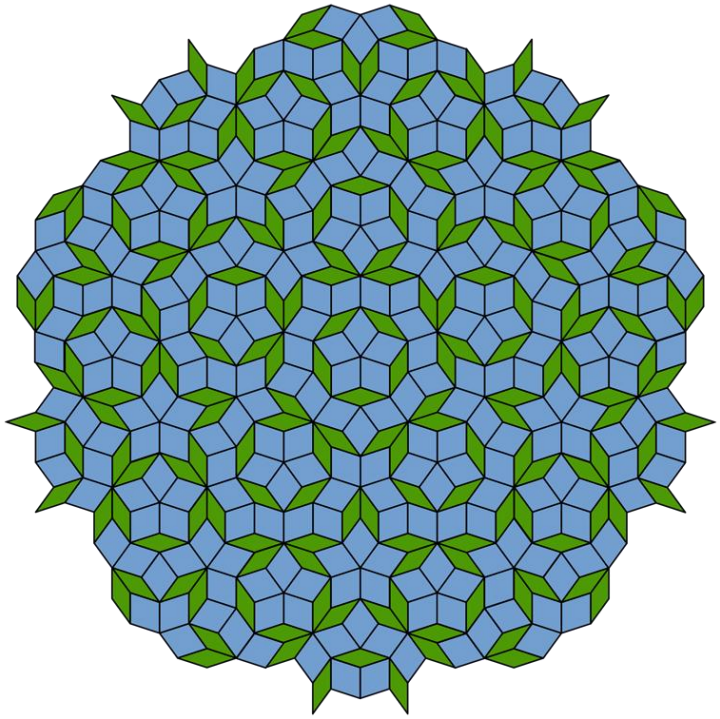
(Image: https://chem.libretexts.org/@go/page/263188)

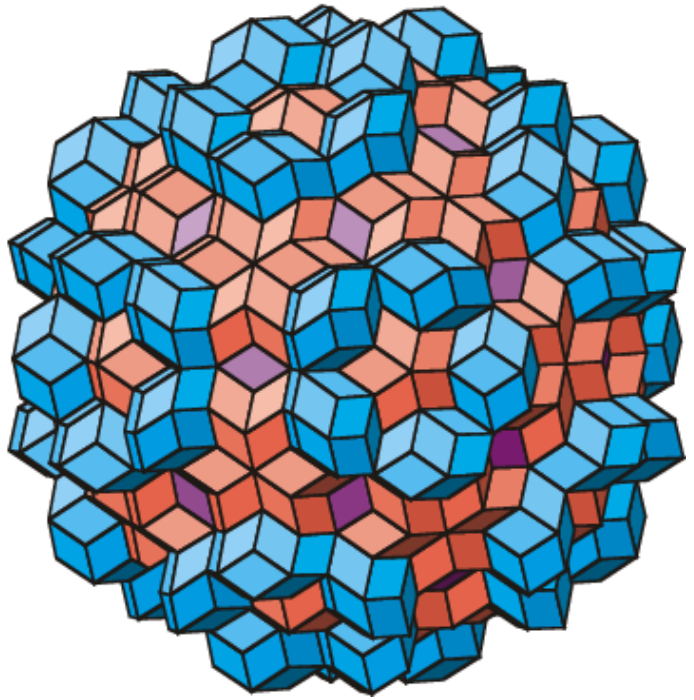
Getting Complicated …



(Image: https://doi.org/10.48550/arXiv.1512.08270)

Quasicrystals



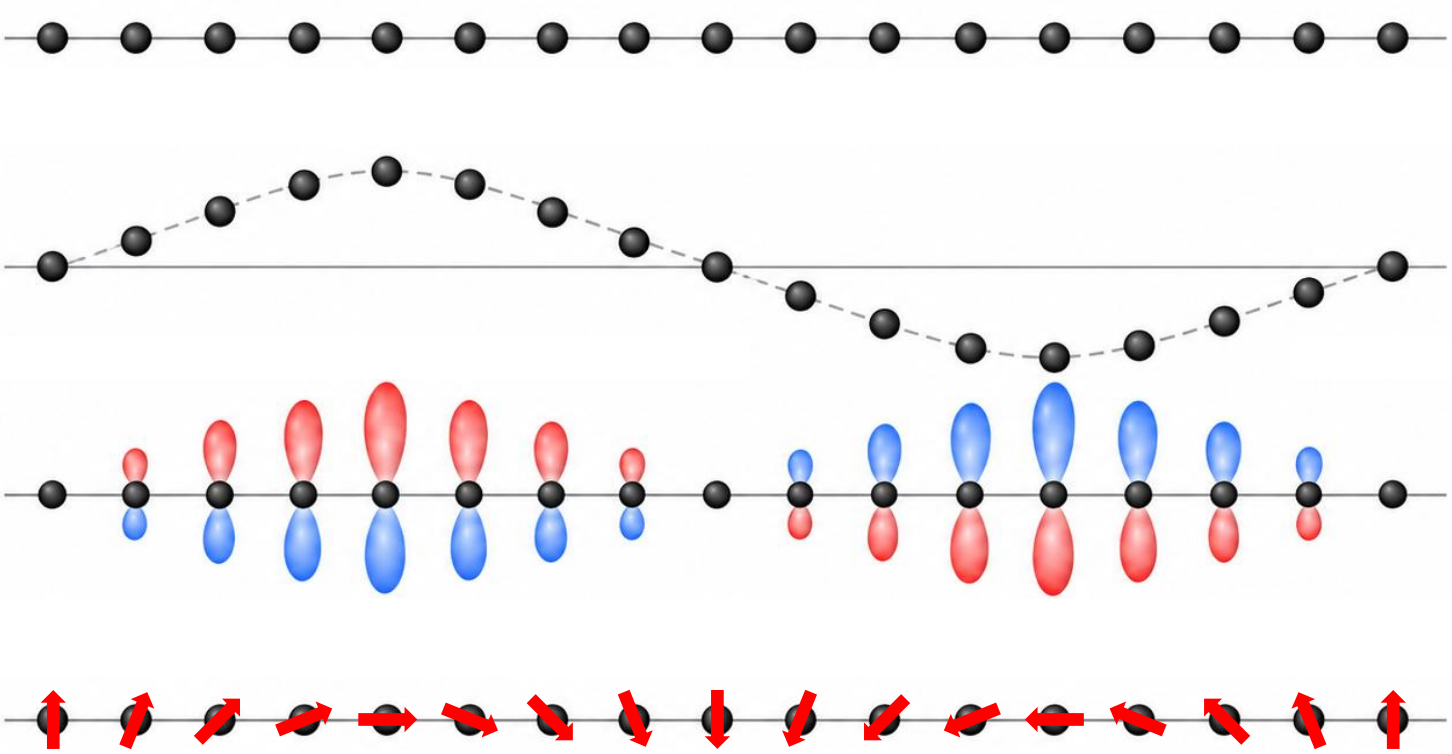




Nobel prize in Chemistry 2011

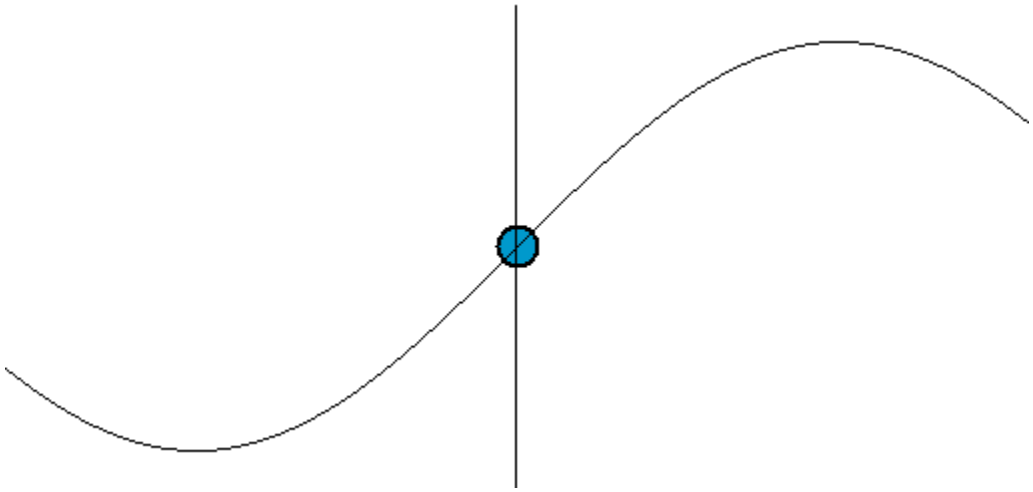
(Image: https://physics.aps.org/story/v28/st14 & Madison, *RSC Advances* 2015)

Periodicity in Crystalline Solids



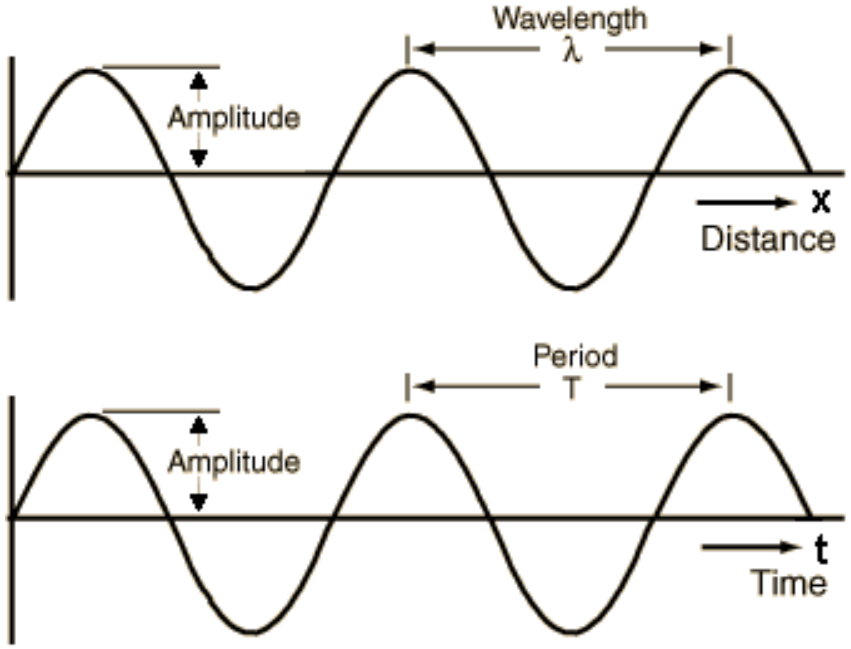
Waves

Temporal change + Spatial change



(Image: https://commons.wikimedia.org/wiki/File:Simple_harmonic_motion_animation.gif)

Basic Terminology



Wavenumber $\tilde{\nu} = 1/\lambda$

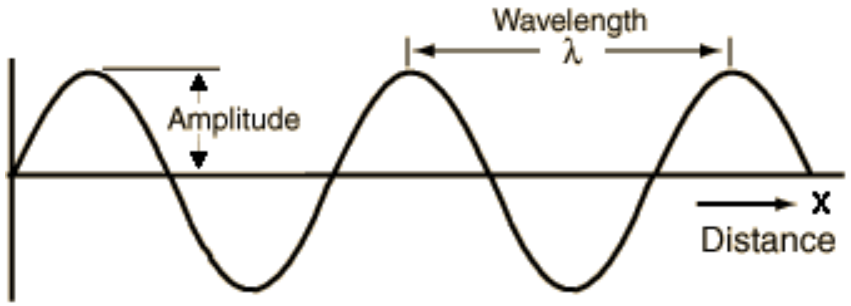
Frequency $\nu = 1/T$

Speed $u = \lambda/T = \lambda\nu$

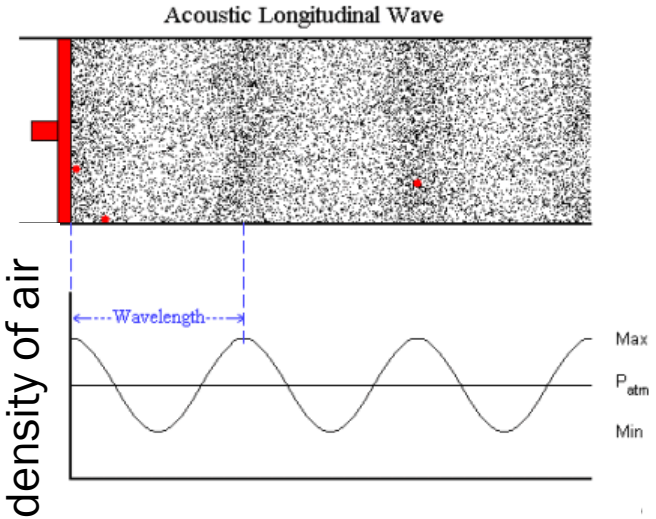
(Image: http://www.sengpielaudio.com/calculator-period.htm)

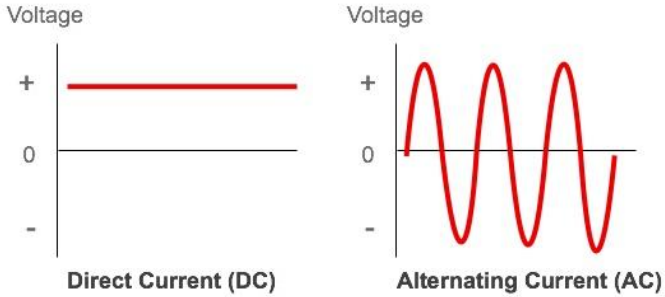
“Amplitude”

Some value assigned for each point of the space

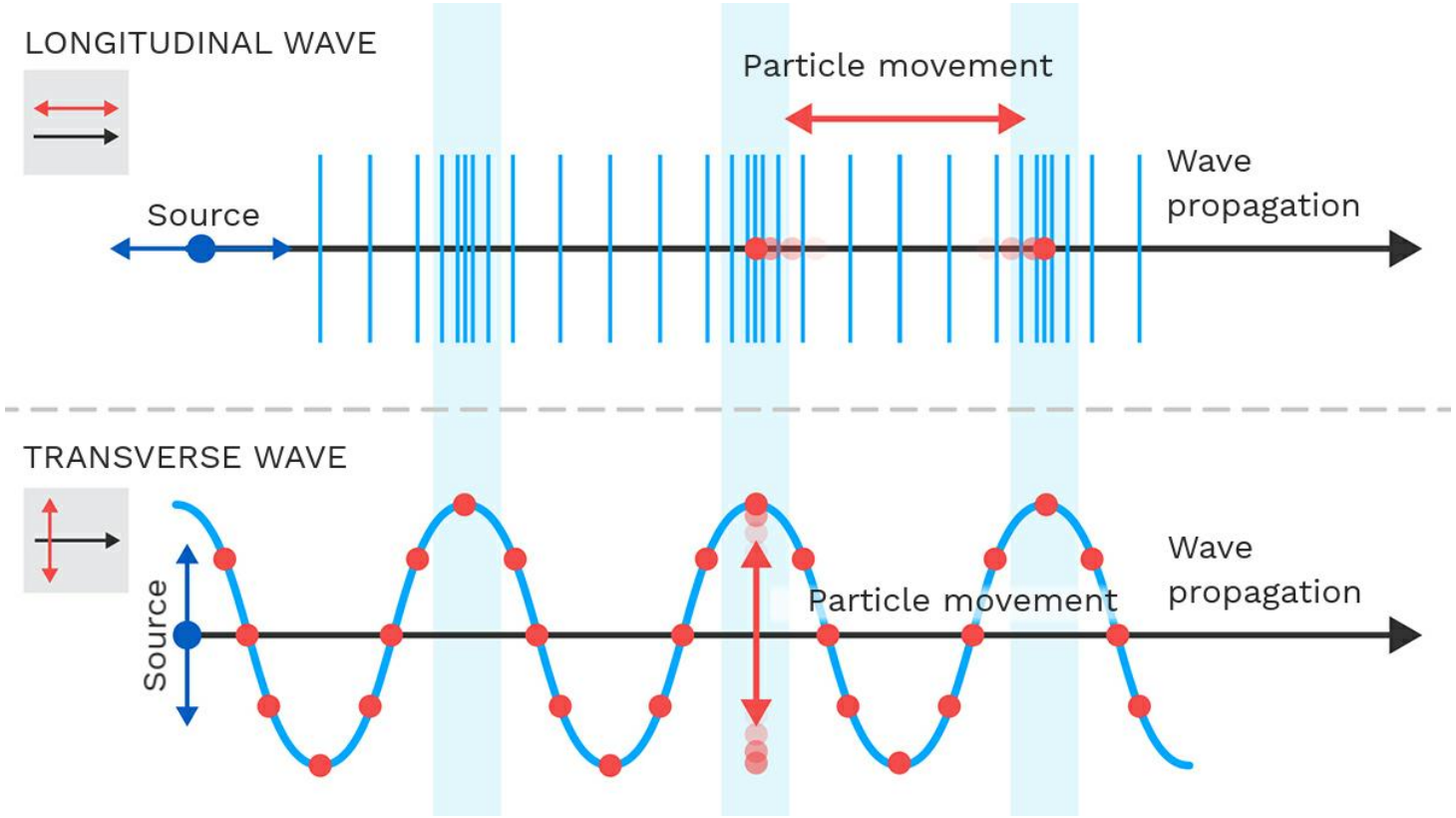


0 +1 +2 +1 0 -1 -2 -1 0 +1 +2 +1





Longitudinal vs. Transverse Waves

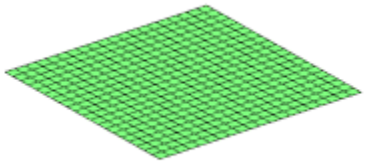


(Image: https://theory.labster.com/longitudinal-transverse-waw/)

Wave Equation

$$\frac{\partial^2 A}{\partial t^2} = u^2 \nabla^2 A$$





(Images: https://upload.wikimedia.org/wikipedia/commons/1/1f/Wave_equation_1D_fixed_endpoints.gif
https://upload.wikimedia.org/wikipedia/commons/b/bb/2D_Wave_Function_resize.gif)

1D Traveling Wave



$$A(x,t) = A_0 \exp(ikx - i\omega t)$$

Wave equation: $\frac{\partial^2 A}{\partial t^2} = u^2 \frac{\partial^2 A}{\partial x^2}$

Substitution leads to

$$-A_0\omega^2 \exp(ikx - i\omega t) = -A_0 u^2 k^2 \exp(ikx - i\omega t)$$

Hence, the wave equation is satisfied with $u = \omega/k$

(Image: https://upload.wikimedia.org/wikipedia/commons/8/89/Travelling_wave_animated_plot.gif)

1D Traveling Wave



$$A(x,t) = A_0 \exp(ikx - i\omega t)$$

- Period: $\omega t = 2\pi$

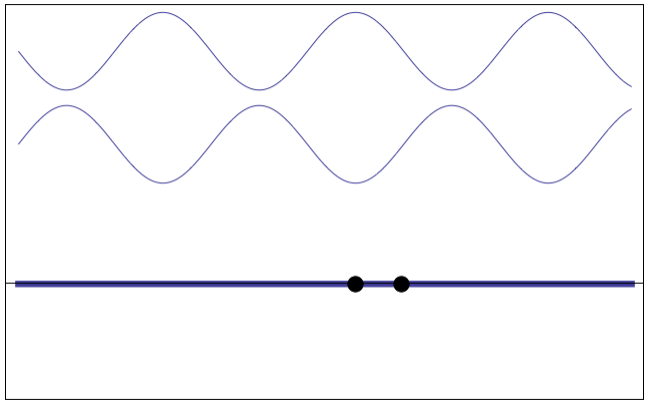
$$T = 2\pi/\omega$$

- Wavelength: $kx = 2\pi$

$$\lambda = 2\pi/k$$

(Image: https://upload.wikimedia.org/wikipedia/commons/8/89/Travelling_wave_animated_plot.gif)

Standing Waves



$$A(x,t) = A_0\cos(kx - \omega t) + A_0\cos(kx + \omega t)$$

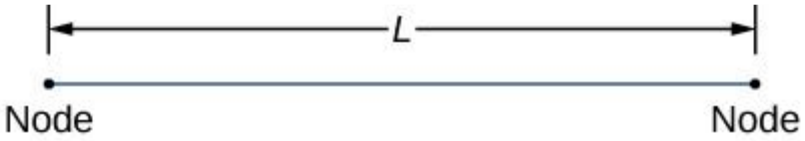
$$= A_0[\cos(kx)\cos(\omega t) + \sin(kx)\sin(\omega t)]$$

$$+ A_0[\cos(kx)\cos(\omega t) - \sin(kx)\sin(\omega t)]$$

$$= 2A_0\cos(kx)\cos(\omega t)$$

(Image: https://www.acs.psu.edu/drussell/demos/superposition/superposition.html)

Angular Wavenumber k



$$k = 2\pi/\lambda$$



$\lambda = 2L$ $\quad k = \pi/L$



$\lambda = L$ $\quad k = 2\pi/L$



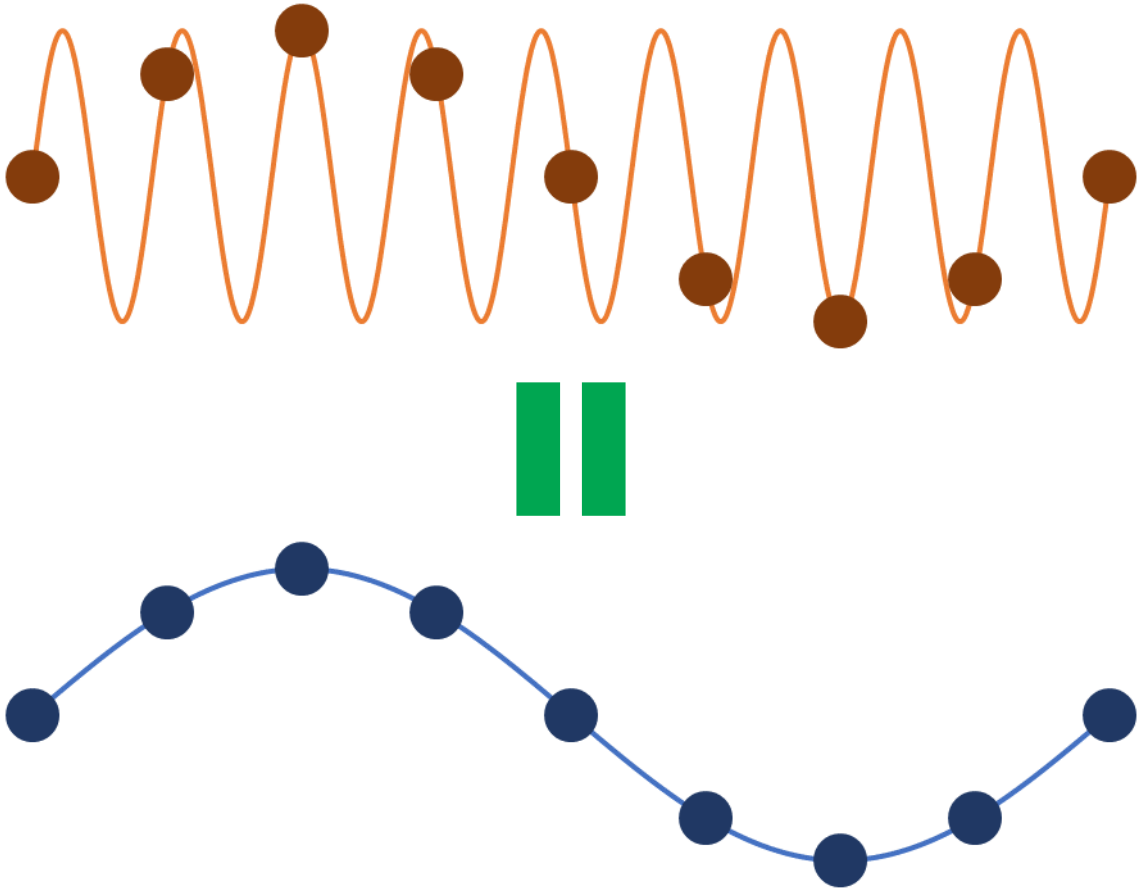
$\lambda = 2L/3$ $\quad k = 3\pi/L$



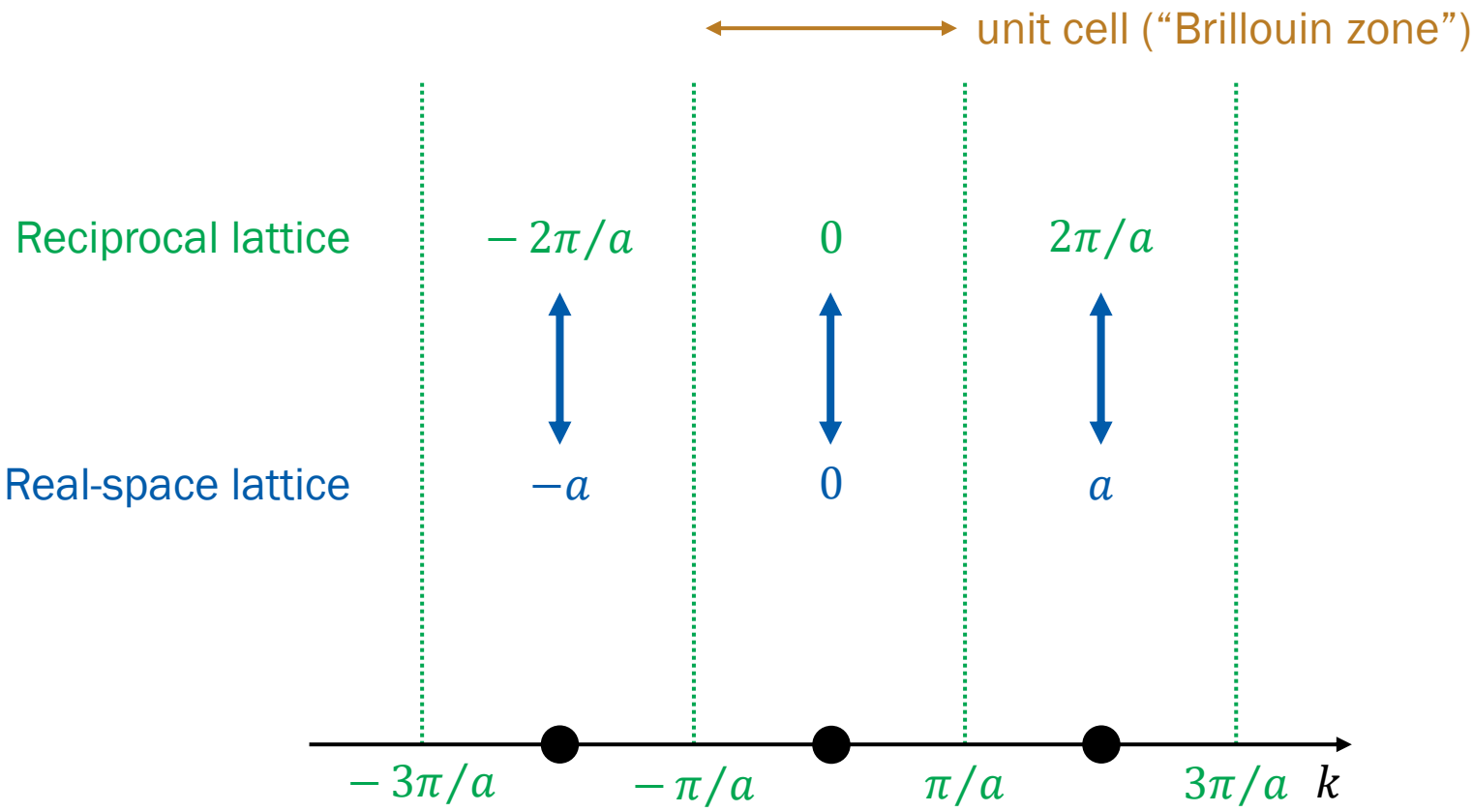
$\lambda = L/2$ $\quad k = 4\pi/L$

(Image: https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/16-6-standing-waves-and-resonance/)

Now Consider Atomic Spacing



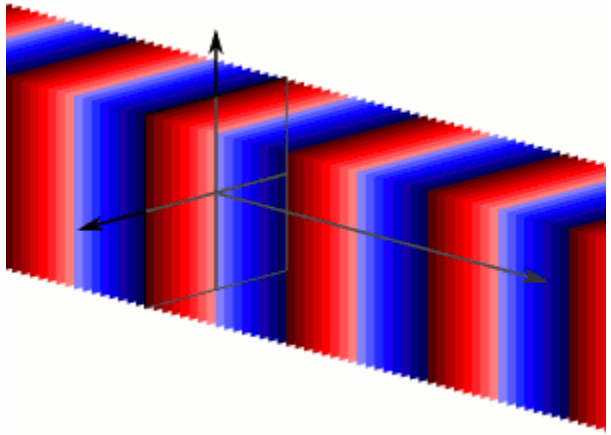
Reciprocal Lattice



3D Generalization

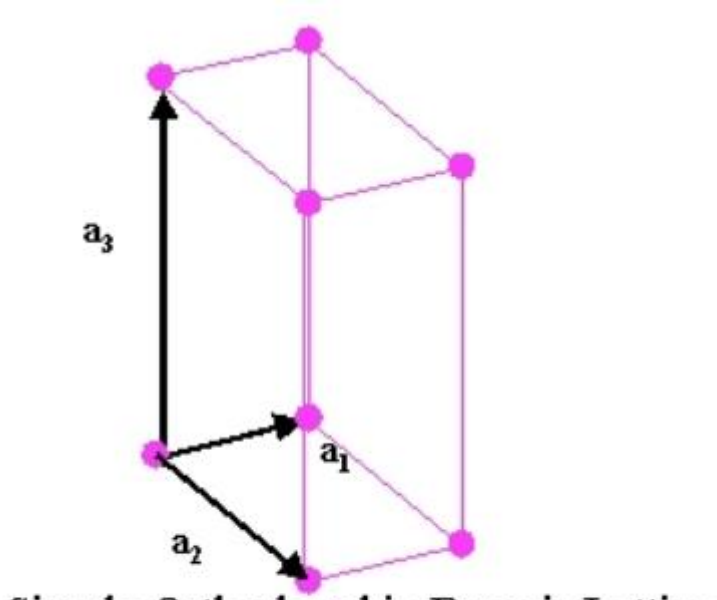


$$A(x,t) = A_0 \exp(ikx - i\omega t)$$

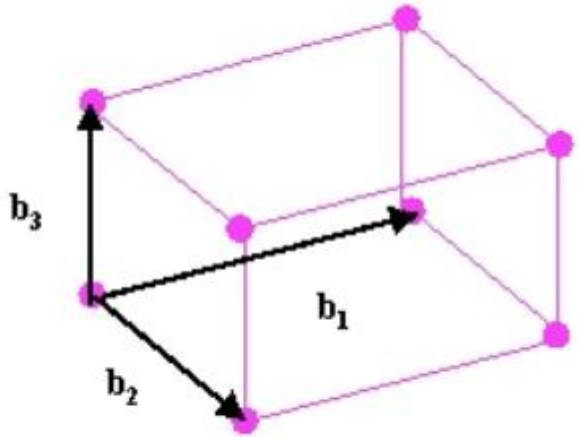


(Images: https://upload.wikimedia.org/wikipedia/commons/8/89/Travelling_wave_animated_plot.gif
https://commons.wikimedia.org/wiki/File:Plane_Wave_3D_Animation_300x216_255Colors.gif)

Reciprocal Lattice: Simple Case



Simple Orthorhombic Bravais Lattice
with $a_3 > a_2 > a_1$



Reciprocal Lattice
Note: $b_1 > b_2 > b_3$

(Image: https://www.physics.smu.edu/scalise/P6338sp16/albert/chapter2.pdf)

Reciprocal Lattice: General Case

For the real-space primitive lattice vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3$, the primitive lattice vectors of the reciprocal lattice must satisfy

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$$

which leads to

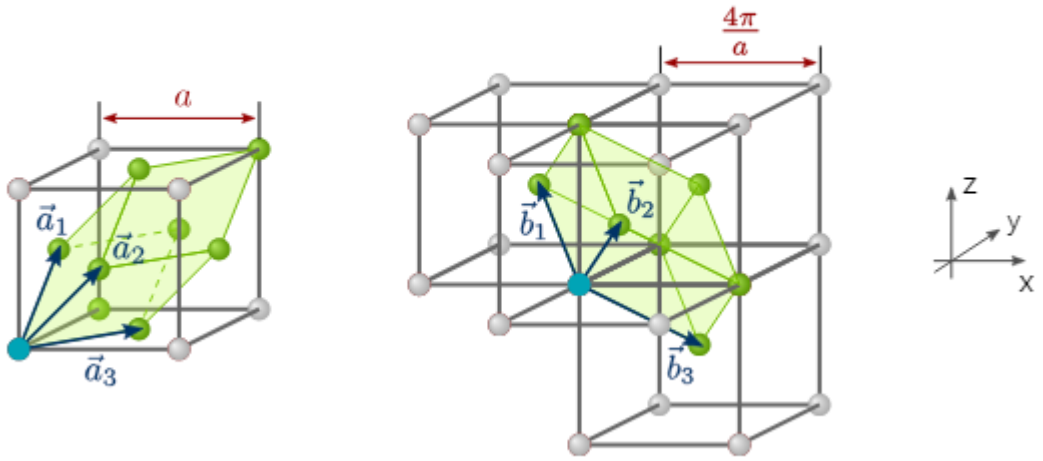
$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}; \quad \vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)};$$

$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

Reciprocal Lattice: Example

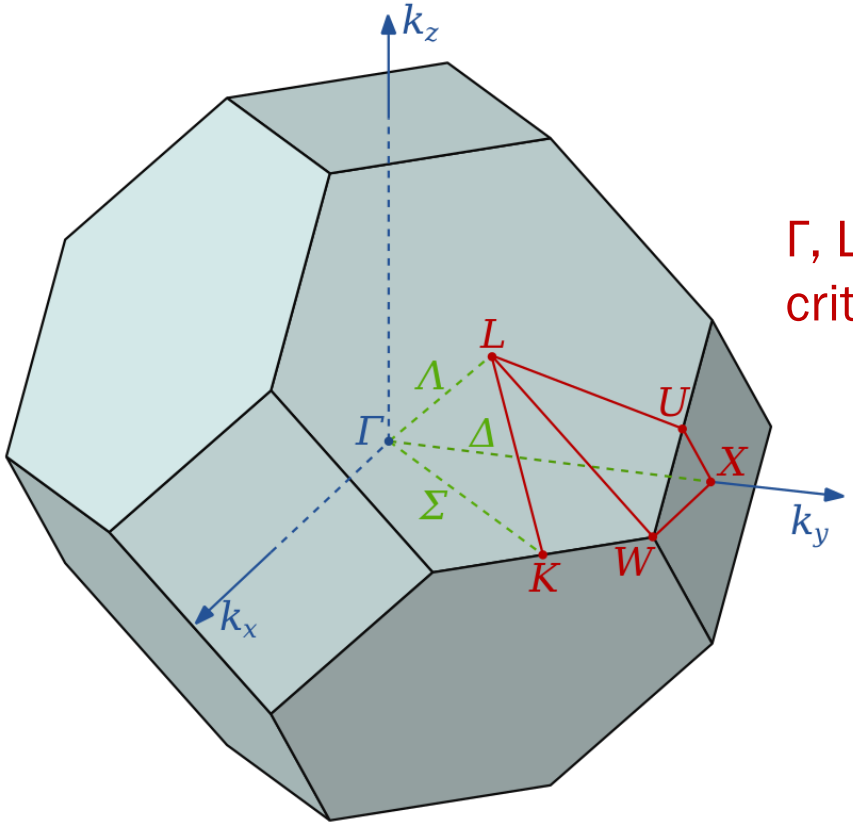
direct lattice:
fcc with edge length a

reciprocal lattice:
bcc with edge length 4π/a



(Image: https://www.physics-in-a-nutshell.com/article/15/the-reciprocal-lattice)

Brillouin Zone (FCC)



Γ, L, U, X, W, K:
critical points

(Image: https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg)